

Lecture 17

The quantum inverse scattering method

Objectives. This is [Lecture 8](#), quantised. The Sklyanin bracket becomes the RTT relation. The transfer matrix becomes an operator that commutes with itself at every value of the spectral parameter. At one special value it is the lattice shift, and the Heisenberg chain's Hamiltonian falls out of its logarithmic derivative, followed by a tower of local charges. Then the lecture restores $\hbar$ in the quantum periodic Toda chain and recovers the r -matrix of [Lecture 10](#) as $\hbar \rightarrow 0$, sign for sign.

17.1 From brackets to commutators

[Lecture 8](#) built commuting integrals from one structure. A chain of N sites has a Lax matrix $L^{(n)}(\lambda)$ at each site, and the site matrices at different sites Poisson-commute (the chain is *ultralocal*). If each obeys the quadratic bracket

$$\{L_1(\lambda), L_2(\mu)\} = [r_{12}(\lambda - \mu), L_1(\lambda)L_2(\mu)], \quad (17.1)$$

then so does the monodromy matrix $T = L^{(N)} \cdots L^{(1)}$ ([Theorem 8.17](#)), and the transfer matrices $t = \text{tr } T$ Poisson-commute for all λ, μ ([Corollary 8.18](#)). The periodic Toda chain is the example: its site matrix ([10.8](#)) satisfies (17.1) with $r(\lambda) = -P/\lambda$ ([Proposition 10.4](#)).

Quantisation makes the entries of L operators and brackets commutators, $\{\cdot, \cdot\} \rightarrow [\cdot, \cdot]/i\hbar$ ([13.1](#)); the legs of [Definition 8.1](#) survive. § 8.9 found the quantum relation that returns (17.1) at order $\hbar$:

$$R_{12}(\lambda - \mu) T_2(\mu) T_1(\lambda) = T_1(\lambda) T_2(\mu) R_{12}(\lambda - \mu), \quad R = \mathbb{1} + i\hbar r + O(\hbar^2), \quad (17.2)$$

which is (8.17). **The order of the two monodromies is not a choice.** With the $+$ in $R = \mathbb{1} + i\hbar r$ fixed by (8.16), the other order returns the bracket with $r \rightarrow -r$ ([Remark 8.19](#)); § 17.6 takes the limit properly. This is the *RTT relation*.

From here to § 17.6, $\hbar = 1$. The auxiliary copies of $\mathbb{C}^2$ get letters, a and b , to keep them apart from the sites $n = 1, \dots, N$ of the chain. The lecture constructs criterion (T) of § 13.5 in full for one model.

17.2 The Lax operator of the Heisenberg chain

The quantum space is the Heisenberg chain of [Lecture 15](#), $\mathcal{H} = (\mathbb{C}^2)^{\otimes N}$, one spin $\frac{1}{2}$ per site. The auxiliary space a is one more $\mathbb{C}^2$. The Lax operator at site n is

$$L_{an}(\lambda) = \lambda + iP_{an} = \left(\lambda + \frac{i}{2}\right) + i\sigma_a \cdot S_n = \begin{pmatrix} \lambda + \frac{i}{2} + iS_n^z & iS_n^- \\ iS_n^+ & \lambda + \frac{i}{2} - iS_n^z \end{pmatrix}_a, \quad (17.3)$$

with P_{an} the permutation of the auxiliary space with site n (Definition 8.3), $\mathbf{S} = \boldsymbol{\sigma}/2$ and $S^\pm = S^x \pm iS^y$. The middle form uses $P = \frac{1}{2}(1 + \boldsymbol{\sigma}_a \cdot \boldsymbol{\sigma}_n)$, the identity behind (13.3); the matrix is in the auxiliary space, with entries operators on site n . Like the Toda site matrix it is linear in the spectral parameter and acts on one site.

Figure 17.1 draws it. Each space is a line: the auxiliary space horizontal, the site vertical. An operator on both is a vertex where they cross, and (17.3) says what happens there: with weight λ the two states go straight through, and with weight i they swap lines, the P of Figure 8.2.

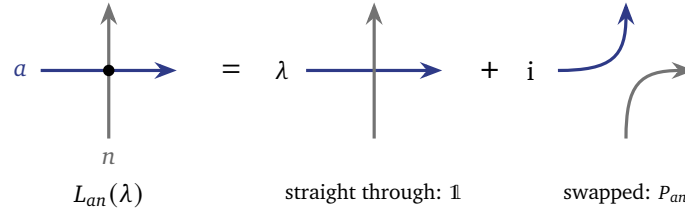


Figure 17.1: The Lax operator as a vertex. The auxiliary line (indigo) crosses the line of site n (grey). With weight λ both states go straight on; with weight i the auxiliary state leaves on the site's line and the site's state on the auxiliary line. (In Figure 8.2, whose lines run side by side, the swap was drawn as a crossing; for lines that meet at right angles the crossing is the identity and the swap is two turns.)

Two auxiliary spaces a, b and one site n make three copies of $\mathbb{C}^2$, and on them the only operators needed are the three transpositions P_{ab}, P_{an}, P_{bn} . Their products are permutations of three objects, and a product of two different transpositions is a 3-cycle, of which there are two:

$$C = P_{ab}P_{an} = P_{an}P_{bn} = P_{bn}P_{ab}, \quad C^{-1} = P_{an}P_{ab} = P_{ab}P_{bn} = P_{bn}P_{an}. \quad (17.4)$$

(Follow a vector $x \otimes y \otimes z$ through each product; this is the computation of Proposition 8.14 and Proposition 16.9.)

Theorem 17.1 (The RLL relation). With $R(\lambda) = \lambda - iP$,

$$R_{ab}(\lambda - \mu) L_{bn}(\mu) L_{an}(\lambda) = L_{an}(\lambda) L_{bn}(\mu) R_{ab}(\lambda - \mu) \quad (17.5)$$

for all λ, μ . With $\lambda + iP$ in place of R it fails.

Proof. Take $R = (\lambda - \mu) + siP_{ab}$ with a sign $s = \pm 1$ and compute the difference Δ of the two sides. It has two parts. The first is $(\lambda - \mu)(L_{bn}L_{an} - L_{an}L_{bn})$, and in $L_{bn}L_{an} - L_{an}L_{bn}$ everything cancels except the products of two P 's: $P_{an}P_{bn} - P_{bn}P_{an} = C - C^{-1}$. The second is

$$si(P_{ab}L_{bn}L_{an} - L_{an}L_{bn}P_{ab}) = si\left(i\mu(P_{ab}P_{an} - P_{an}P_{ab}) + i\lambda(P_{ab}P_{bn} - P_{bn}P_{ab}) - (P_{ab}P_{bn}P_{an} - P_{an}P_{bn}P_{ab})\right),$$

the terms $\lambda\mu P_{ab}$ having cancelled. By (17.4) the first bracket is $C - C^{-1}$, the second $C^{-1} - C$, and the triple products are both P_{bn} , so they cancel. The second part is therefore $s(\lambda - \mu)(C - C^{-1})$, and

$$\Delta = (1 + s)(\lambda - \mu)(C - C^{-1}).$$

This vanishes for $s = -1$ and is not zero for $s = +1$, since $C \neq C^{-1}$. ■

The whole relation is a statement about the permutations of three objects (the check script confirms the residual symbolically).

Remark 17.2 (A warning for readers of other books). Most of the literature writes the RTT relation the other way round, and each book pairs it with its own R and its own Lax operator. Faddeev (Les Houches 1995, §3) uses $R'(\lambda) = \lambda + iP$ and $R'_{ab}(\lambda - \mu) T_a(\lambda) T_b(\mu) = T_b(\mu) T_a(\lambda) R'_{ab}(\lambda - \mu)$, with a Lax operator $(\lambda - \frac{i}{2}) + iP$. *This is the same relation as ours:* exchange (a, λ) with (b, μ) in (17.7) and use $R(-u) = -R'(u)$; his Lax operator is ours at $\lambda - \frac{i}{2}$. Korepin, Bogoliubov and Izergin (§§VI.1, VI.5) use the same ordering as Faddeev but with $\bar{R} = \mathbb{1} - iP/(\lambda - \mu)$ and the Lax operator $(\lambda + \frac{i}{2}) - iP$, whose spin term has the opposite sign; their monodromy is ours up to a sign at a reflected argument, $T(\lambda) = (-1)^N T_{\text{KBI}}(-\lambda - \frac{i}{2})$. Nothing in those books is wrong. What is wrong is to mix them: copying $\lambda + iP$ into this course's ordering gives a false relation (Theorem 17.1), and every sign downstream, in particular every exchange relation of Lecture 18, then disagrees with these notes. **Rule: before using another book's formula, find its ordering, its R and its Lax operator, and convert all three; never one without the others.** For Faddeev the classical r is minus ours, which is Remark 8.19 again.

17.3 From one site to the chain

The monodromy matrix

The *monodromy matrix* of the chain is the ordered product of the Lax operators along it,

$$T_a(\lambda) = L_{aN}(\lambda) \cdots L_{a2}(\lambda) L_{a1}(\lambda) = \begin{pmatrix} A(\lambda) & B(\lambda) \\ C(\lambda) & D(\lambda) \end{pmatrix}_a, \quad (17.6)$$

an operator on $\mathbb{C}_a^2 \otimes \mathcal{H}$: in the auxiliary space a 2×2 matrix whose entries A, B, C, D are operators on the whole chain. As a picture it is a row: one auxiliary line crossing every site in turn (Figure 17.2), Figure 8.5 with operators at the vertices. Operators on different sites commute, so L_{an} commutes with L_{bm} for $n \neq m$: they act on disjoint sets of spaces. That is the quantum form of ultralocality, and it is all the next proof uses.

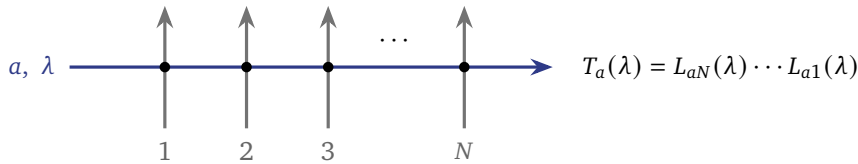


Figure 17.2: The monodromy matrix as a row. The auxiliary line meets site 1 first, so L_{a1} is the rightmost factor. The two free ends of the auxiliary line are the two indices of the 2×2 matrix (17.6); the ends of the site lines are the quantum space.

Theorem 17.3 (RLL implies RTT). *The monodromy (17.6) satisfies*

$$R_{ab}(\lambda - \mu) T_b(\mu) T_a(\lambda) = T_a(\lambda) T_b(\mu) R_{ab}(\lambda - \mu). \quad (17.7)$$

The same holds for any chain whose Lax operators obey (17.5) with a common R and act on different sites.

Proof. In $T_b T_a = L_{bN} \cdots L_{b1} L_{aN} \cdots L_{a1}$, move each L_{an} to the left past $L_{b,n-1}, \dots, L_{b1}$, with which it commutes. The factors pair up by site:

$$T_b(\mu) T_a(\lambda) = (L_{bN} L_{aN}) (L_{b,N-1} L_{a,N-1}) \cdots (L_{b1} L_{a1}).$$

Now R_{ab} enters on the left. By (17.5) at site N , $R_{ab} L_{bN} L_{aN} = L_{aN} L_{bN} R_{ab}$: it has passed the first pair and reversed it. The same step at each site in turn carries R_{ab} to the right end, leaving $(L_{aN} L_{bN}) \cdots (L_{a1} L_{b1}) R_{ab}$, and undoing the pairing gives $T_a T_b R_{ab}$. ■

This is the *train argument* (Figure 17.3): the crossing of the two auxiliary lines is pushed through the row one site at a time, as the bracket was pushed through the product by Leibniz's rule in Theorem 8.17.

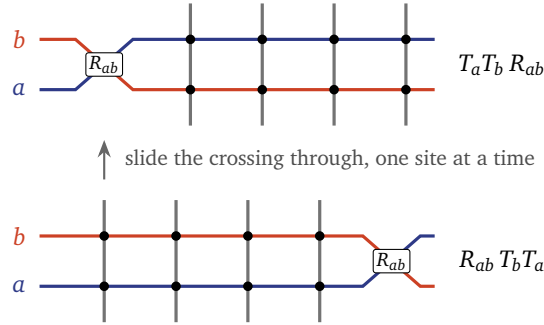


Figure 17.3: The train argument. Operators act left to right along the auxiliary lines, so the leftmost thing drawn is the rightmost factor. Bottom: the rows, then the crossing R_{ab} . By (17.5) the crossing slides through site N , then $N - 1, \dots$, until it has passed the row (top): (17.7).

The Lax operator is an R -matrix, and RLL is the Yang–Baxter equation

Compare (17.3) with R :

$$L_{an}(\lambda) = \lambda + iP_{an} = -R_{an}(-\lambda). \quad (17.8)$$

The Lax operator is the R -matrix itself, with the site as its second space. It is *not* $R_{an}(\lambda)$: the two differ by $\lambda \rightarrow -\lambda$, and confusing them is the trap of Remark 17.2 in another guise. Substitute (17.8) into (17.5); the two minus signs cancel,

$$R_{ab}(\lambda - \mu) R_{bn}(-\mu) R_{an}(-\lambda) = R_{an}(-\lambda) R_{bn}(-\mu) R_{ab}(\lambda - \mu).$$

Rename $b \rightarrow 1, a \rightarrow 2, n \rightarrow 3$, and put $u = -\mu, v = -\lambda$, so that $\lambda - \mu = u - v$. Since $P_{12} = P_{21}$ this is

$$R_{12}(u - v) R_{13}(u) R_{23}(v) = R_{23}(v) R_{13}(u) R_{12}(u - v), \quad (17.9)$$

the *Yang–Baxter equation* in its standard form. With $u = u_{13}, v = u_{23}$ and $u - v = u_{12}$ it is the equation (16.17) that Theorem 16.8 derived from three-body consistency, for the matrix $X_B(u) = R(u)/(u + i)$ of the delta bosons (16.14). So Theorem 17.1 and Proposition 16.9 are one fact, read twice. In Lecture 16 the three lines were three particles and R a two-body collision; in the Heisenberg chain one of the three lines is a site. The fermions of Lecture 16 scatter with $X_F = -L(u)/(u + i)$, the Lax operator itself.

Figure 17.5 shows the equation as a picture. It is Figure 8.4 and Figure 16.6, and Figure 17.4 is its special case with one line vertical.

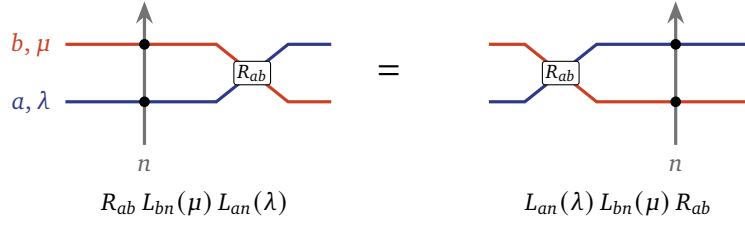


Figure 17.4: The RLL relation (17.5). Left: the site line n meets a (lower) and then b , and afterwards the auxiliary lines cross. Right: they cross first, and n meets b and then a . The crossing of a and b may be slid through the site line.

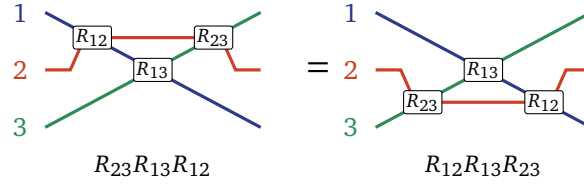


Figure 17.5: The Yang-Baxter equation (17.9) as a slide: line 2 passes either above or below the crossing of lines 1 and 3. Lines carry spectral parameters and a crossing of lines i, j carries R_{ij} at the difference of their parameters, which is why $u_{13} = u_{12} + u_{23}$. Reading left to right, the first crossing is the rightmost factor. Make line 3 a site and this is Figure 17.4.

17.4 Commuting transfer matrices

The *transfer matrix* is the trace of the monodromy over the auxiliary space,

$$t(\lambda) = \text{tr}_a T_a(\lambda) = A(\lambda) + D(\lambda), \quad (17.10)$$

an operator on the chain alone: in the picture the auxiliary line is closed into a loop (Figure 17.6). It is a polynomial of degree N in λ with operator coefficients, and leading term $2\lambda^N$.

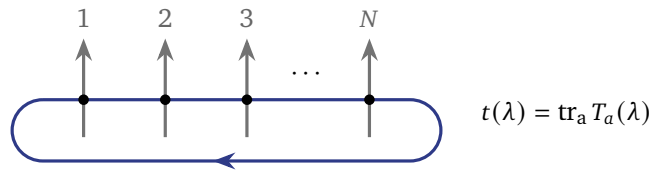


Figure 17.6: The transfer matrix: the auxiliary line of Figure 17.2, closed into a loop by the trace. The loop has no ends, so it acts on the sites alone; this is Figure 8.6 with operators at the vertices.

Theorem 17.4 (Commuting transfer matrices). $[t(\lambda), t(\mu)] = 0$ for all $\lambda, \mu \in \mathbb{C}$.

Proof. $R(u)R(-u) = (u - iP)(-u - iP) = -(u^2 + 1)$, so $R(u)$ is invertible for $u \neq \pm i$. For such $\lambda - \mu$, multiply (17.7) on the left by R_{ab}^{-1} : $T_b(\mu)T_a(\lambda) = R_{ab}^{-1}T_a(\lambda)T_b(\mu)R_{ab}$. Take the trace over both auxiliary spaces. On the left, $\text{tr}_{ab} T_b T_a = t(\mu)t(\lambda)$, because T_a and T_b have their auxiliary legs in different spaces. On the right, R_{ab} acts on the auxiliary spaces only, as a numerical matrix, and the partial trace is cyclic for such a factor; so $\text{tr}_{ab} (R_{ab}^{-1}T_a T_b R_{ab}) = \text{tr}_{ab} (T_a T_b) = t(\lambda)t(\mu)$. Both sides are polynomials in λ, μ , so the identity extends to $\lambda - \mu = \pm i$. ■

This is [Corollary 8.18](#): there the trace of a commutator vanished, here the trace of a conjugate is the trace ([Figure 17.7](#)). The coefficients of $t(\lambda)$ commute with one another. Whether the family contains a Hamiltonian worth having, and whether its members are local, is the business of § 17.5.

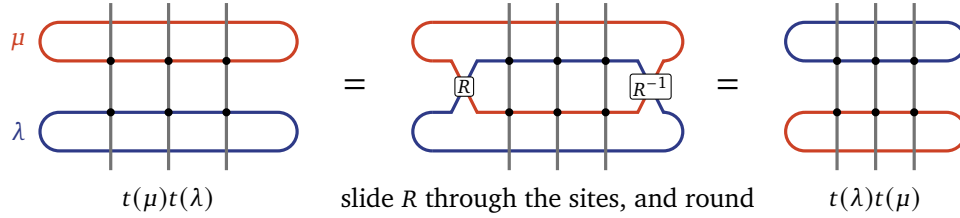


Figure 17.7: Why transfer matrices commute. Insert $RR^{-1} = \mathbb{1}$ between the loops; R slides through every site ([Figure 17.3](#)), exchanging the rows, and meets R^{-1} round the back. The loops have passed through each other.

What RTT gives, and what it does not

The proof used (17.7) and the invertibility of R , and not the Yang–Baxter equation. The equation does two other jobs. It *supplies Lax operators*: by (17.8) the rational R , read with a site as its second space, is a Lax operator for its own RLL relation, which is why the Heisenberg chain has one. (The step used $R_{12} = R_{21}$; for a solution without that symmetry the recipe is $L_{an}(\lambda) = -R_{na}(-\lambda)$.) And it is *sufficient, not forced*, for the consistency of the RTT relations when they are taken to define an algebra: reordering three monodromies along the two routes of [Figure 17.5](#) requires a combination of R ’s only to commute with the triple products — [Remark 8.13](#) quantised, proved in the reading at the end. This is [Lecture 0](#)’s “sufficient, not forced”. For the Heisenberg chain nothing is left to choose: the check script finds that (17.5) fixes R up to a scalar for one and two sites, at generic λ, μ .

Finally, *integrability does not require RTT* : the free-fermion charges of [Lecture 14](#) were constructed directly, and the Calogero–Sutherland charges of [Lecture 21](#) will be too. Nor does RTT decide completeness, which remains [Lecture 15](#)’s cited question.

17.5 The regular point: the shift, the Hamiltonian, the tower

At $\lambda = 0$ the transfer matrix is the shift

At $\lambda = 0$ the Lax operator is a pure swap, $L_{an}(0) = iP_{an}$, and every vertex of the loop turns ([Figure 17.8](#)). The state that enters the loop is deposited on site 1, while site 1’s old state is carried along the auxiliary line to site 2; the loop passes each site’s state on to the next, and the trace brings the last one round to site 1.

Proposition 17.5 (The shift). $t(0) = i^N U$, where U is the cyclic shift, $U(v_1 \otimes v_2 \otimes \cdots \otimes v_N) = v_N \otimes v_1 \otimes \cdots \otimes v_{N-1}$: the state of site n moves to site $n+1$. Explicitly $U = P_{N,N-1} P_{N,N-2} \cdots P_{N,1}$. On a magnon of momentum k (15.4), $U = e^{-ik}$.

Proof. $T_a(0) = i^N P_{aN} P_{a,N-1} \cdots P_{a1}$. The identity $P_{aN} P_{am} = P_{Nm} P_{aN}$ (the first equality chain of (17.4), relabelled) moves P_{aN} to the right end one factor at a time: $P_{aN} \cdots P_{a1} = P_{N,N-1} \cdots P_{N,1} P_{aN}$. Now $\text{tr}_a P_{aN} = \mathbb{1}$, since $P = \sum_{ij} E_{ij} \otimes E_{ji}$ and $\text{tr} E_{ij} = \delta_{ij}$. Applying

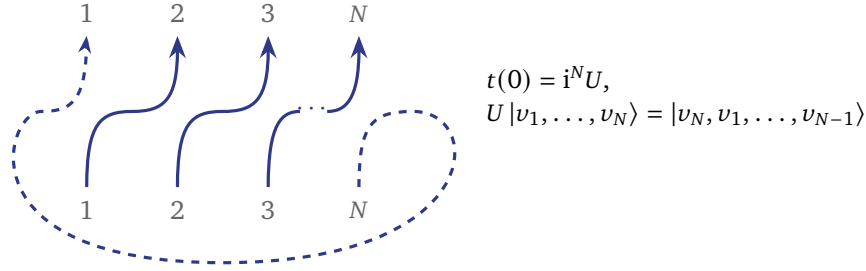


Figure 17.8: The regular point. At $\lambda = 0$ every vertex is a swap: the state arriving from below on site n turns onto the auxiliary line and leaves upward on site $n + 1$. The dashed arc is the trace, which carries site N 's state round to site 1. The loop has become the cyclic shift.

$P_{N,1}$, then $P_{N,2}, \dots$, to $v_1 \otimes \dots \otimes v_N$ puts v_N on site 1, v_1 on site 2, and so on. On $|x\rangle$, a down spin at x , $U|x\rangle = |x+1\rangle$, so $U \sum_x e^{ikx} |x\rangle = e^{-ik} \sum_x e^{ikx} |x\rangle$. ■

So the first member of the family is the lattice translation, and the total momentum K is read off from $U = e^{-iK}$. Since $t(0)$ is invertible, the operator $t(0)^{-1}t(\lambda) = \mathbb{1} + O(\lambda)$ has a logarithm near $\lambda = 0$, a power series in λ whose coefficients commute with every $t(\mu)$. Write $\frac{d^k}{d\lambda^k} \ln t(\lambda)|_0$ for $k!$ times its λ^k coefficient; for $k = 1$ it is $t(0)^{-1}t'(0)$.

The Hamiltonian

Lemma 17.6 (One vertex straightened). *Delete the factor P_{am} from $P_{aN} \cdots P_{a1}$ and trace over a . The result is $U P_{m-1,m}$, with $P_{0,1} := P_{N,1}$ closing the ring.*

Proof (★). Let $F = P_{aN} \cdots P_{a1}$ and let F_m be F without P_{am} . For $m \geq 2$, $P_{am}P_{a,m-1} = P_{a,m-1}P_{m-1,m}$ (both are the same 3-cycle, (17.4)), and $P_{m-1,m}$ commutes with every P_{ak} , $k < m - 1$. So $F = F_m P_{m-1,m}$, hence $F_m = F P_{m-1,m}$ and $\text{tr}_a F_m = U P_{m-1,m}$. For $m = 1$, F_1 is the product for the sites $2, \dots, N$ alone, whose trace is the cyclic shift of those sites, and that is $U P_{N,1}$. ■

Proposition 17.7 (The Hamiltonian from the transfer matrix).

$$\frac{d}{d\lambda} \ln t(\lambda)|_0 = t(0)^{-1}t'(0) = -i \sum_{n=1}^N P_{n,n+1}, \quad \text{so} \quad H = \frac{J}{2} \left(i \frac{d}{d\lambda} \ln t(\lambda)|_0 - N \right) \quad (17.11)$$

is the Heisenberg Hamiltonian (15.1), and it commutes with $t(\mu)$ for every μ .

Proof. $L'_{an} = \mathbb{1}$, so by Leibniz $t'(0)$ is the sum over m of the product with the m -th factor replaced by $\mathbb{1}$: $t'(0) = i^{N-1} \sum_m \text{tr}_a F_m = i^{N-1} U \sum_m P_{m-1,m}$ by Lemma 17.6. Multiply by $t(0)^{-1} = i^{-N} U^{-1}$. Then use $H = \frac{J}{2} \sum (P_{n,n+1} - 1)$. ■

This is the promise of Lecture 8: the expansion that gave the classical integrals of the Toda chain gives the quantum Hamiltonian of the Heisenberg chain. The Heisenberg chain has criterion (T) of § 13.5: its Hamiltonian sits in a commuting family generated by one transfer matrix.

The tower

Why the *logarithm*? Each derivative at 0 straightens one vertex, which after U^{-1} leaves a bond $P_{m-1,m}$ (Lemma 17.6). Two straightened vertices leave two bonds: far apart, a product of two local operators, which is not local; neighbours, one operator on three sites ($P_{m,m+1}P_{m-1,m}$, the check script). The square of the first derivative contains the same far-apart products, and the logarithm cancels them; what is left comes from neighbours and from $P^2 = \mathbb{1}$. For $N \geq 4$

$$\frac{d^2}{d\lambda^2} \ln t(\lambda) \Big|_0 = N + \sum_{n=1}^N [P_{n,n+1}, P_{n+1,n+2}] = N - \frac{i}{2} Q_3, \quad (17.12)$$

with Q_3 the range-3 charge of Theorem 13.4, recovered here without the commutator computation · *sketched; verified at $N = 6, 7, 8$.*

In general, $\frac{d^k}{d\lambda^k} \ln t|_0$ is a sum over the ring of one density of range $k+1$, the same density for every N large enough: a local charge in the sense of Definition 13.2, with range growing with k · *sketched above; proved by Grabowski and Mathieu 1995; the check script verifies range and N -independence for $k \leq 3$ at $N = 8, 9$; they commute by Theorem 17.4.* This is the tower of § 13.3. It rests on the *regular point*, where the Lax operator is a permutation, $t(0)$ is invertible and every vertex only passes states along; at a generic λ , $t(\lambda)$ couples all N sites at once.

17.6 The classical limit, verified: the quantum Toda chain

The Heisenberg chain has no classical limit worth the name — a spin $\frac{1}{2}$ has no small parameter. To test the dictionary, quantise a chain of Lecture 10. Restore $\hbar$. The periodic Toda chain has canonical pairs with $[q_m, p_n] = i\hbar \delta_{mn}$, represented by $p_n = -i\hbar \partial/\partial q_n$ on functions of q . Keep the site matrix of (10.8) as it stands,

$$L_n(\lambda) = \begin{pmatrix} \lambda - p_n & e^{-q_n} \\ -e^{q_n} & 0 \end{pmatrix}, \quad (17.13)$$

now with operator entries. No ordering question arises: each entry involves q_n or p_n , never both.

Theorem 17.8 (Quantum Toda: RLL). *The site matrix (17.13) satisfies*

$$R_{ab}(\lambda - \mu) L_b(\mu) L_a(\lambda) = L_a(\lambda) L_b(\mu) R_{ab}(\lambda - \mu), \quad R(\lambda) = \lambda - i\hbar P,$$

exactly, to all orders in $\hbar$, and fails with $\lambda + i\hbar P$. Consequently the monodromy $T = L_N \cdots L_1$ satisfies RTT, and $[t(\lambda), t(\mu)] = 0$.

Proof. Write $L(\lambda) = \lambda E_{11} + L^0$ with $L^0 = L(0)$. In the leg notation the $(ij), (kl)$ entry of $L_a(\lambda)L_b(\mu)$ is $L_{ik}(\lambda)L_{jl}(\mu)$, and of $L_b(\mu)L_a(\lambda)$ it is the same product in the other order. So their difference is the matrix of commutators $-[L_{ik}, L_{jl}]$, which do not depend on λ, μ . The only non-zero ones are $[p, e^{\mp q}] = \pm i\hbar e^{\mp q}$, exactly $i\hbar$ times the Poisson brackets used in the proof of Proposition 10.4; that proof computed the whole matrix of brackets, so

$$L_b(\mu)L_a(\lambda) - L_a(\lambda)L_b(\mu) = -i\hbar (E_{11} \otimes L^0 - L^0 \otimes E_{11})P.$$

Next, $P L_b(\mu)L_a(\lambda) P = L_a(\mu)L_b(\lambda)$ (conjugating by P exchanges the legs), and expanding $L = \lambda E_{11} + L^0$ gives $L_a(\mu)L_b(\lambda) - L_a(\lambda)L_b(\mu) = (\mu - \lambda)(E_{11} \otimes L^0 - L^0 \otimes E_{11})$; the $L^0 \otimes L^0$ terms,

in the same order on both sides, cancel. With $R = (\lambda - \mu) + s\hbar P$ the difference of the two sides of the relation is

$$(\lambda - \mu)(L_b L_a - L_a L_b) + s\hbar(PL_b L_a - L_a L_b P) = -(1 + s)\hbar(\lambda - \mu)(E_{11} \otimes L^0 - L^0 \otimes E_{11})P,$$

using $PL_b L_a = L_a(\mu)L_b(\lambda)P$. It vanishes for $s = -1$ only. Sites commute, so [Theorem 17.3](#) and [Theorem 17.4](#) apply verbatim, now with $R(u)R(-u) = -(u^2 + \hbar^2)$. ■

The coefficients of $t(\lambda) = \sum_k c_k \lambda^{N-k}$ are the quantum integrals. They are the expressions of (10.14), now as operators, with no ordering ambiguity since each term has at most one factor per site: $c_1 = -\sum p_n$, and $H = \frac{1}{2}c_1^2 - c_2 = \frac{1}{2}\sum p_n^2 + \sum e^{q_n - q_{n+1}}$ is the quantum periodic Toda Hamiltonian. So the quantum chain has a commuting family from one transfer matrix: $c_1, \dots, c_N$ commute, and H is a function of them. (Their independence, and a local tower in the sense of criterion (T), are not shown here.) Its spectral problem was reduced to one-variable quantisation conditions, for small N by Gutzwiller (1981) and for all N by Sklyanin's separation of variables (1985) · *cited*.

$\hbar \rightarrow 0$

To compare with [Lecture 10](#), divide by $\lambda - \mu$:

$$\frac{R(\lambda - \mu)}{\lambda - \mu} = \mathbb{1} + i\hbar r(\lambda - \mu), \quad r(\lambda) = -\frac{P}{\lambda},$$

exactly the r of [Proposition 10.4](#), and (8.16) with no $O(\hbar^2)$ term. The limit needs a rule for turning operators into functions.

Lemma 17.9 (Symbols). Write an operator polynomial in the p 's with all p 's on the right, $F = \sum_\alpha f_\alpha(q) p^\alpha$, and call $\sigma(F) = \sum_\alpha f_\alpha(q) p^\alpha$, a function of (q, p) , its symbol. Then

$$\sigma(FG) = \sum_\alpha \frac{(-i\hbar)^{|\alpha|}}{\alpha!} \partial_p^\alpha \sigma(F) \partial_q^\alpha \sigma(G), \quad \text{so} \quad \sigma([F, G]) = i\hbar \{\sigma(F), \sigma(G)\} + O(\hbar^2).$$

Proof (★). Move each $p_n = -i\hbar \partial_{q_n}$ in F to the right through G 's functions of q by Leibniz's rule, $p_n^k g = \sum_j \binom{k}{j} (-i\hbar)^j (\partial_{q_n}^j g) p_n^{k-j}$. The first-order terms of $\sigma(FG) - \sigma(GF)$ are $-i\hbar \sum_n (\partial_{p_n} f \partial_{q_n} g - \partial_{p_n} g \partial_{q_n} f) = i\hbar \{f, g\}$ in the course's convention. ■

Corollary 17.10 (The classical limit of RTT). The symbols of the entries of the quantum monodromy are the entries of the classical monodromy (10.13). At order $\hbar$ the RTT relation of [Theorem 17.8](#) becomes

$$\{T_1(\lambda), T_2(\mu)\} = [r(\lambda - \mu), T_1(\lambda)T_2(\mu)], \quad r(\lambda) = -\frac{P}{\lambda} :$$

[Proposition 10.4](#) and [Theorem 8.17](#), sign for sign.

Proof. Each entry of T is a sum of products of entries of different site matrices, which commute, so its symbol is the classical product. Write RTT as $T_a T_b - T_b T_a = i\hbar(r T_b T_a - T_a T_b r)$ and take symbols. On the left the $(ij), (kl)$ entry is the commutator $[T_{ik}(\lambda), T_{jl}(\mu)]$, whose symbol is $i\hbar \{T_{ik}, T_{jl}\} + O(\hbar^2)$ by [Lemma 17.9](#): the entry of $i\hbar \{T_1, T_2\}$ in [Definition 8.1](#). On the right the symbols of $T_b T_a$ and $T_a T_b$ are both $T_1 T_2 + O(\hbar)$. Divide by $i\hbar$ and let $\hbar \rightarrow 0$. ■

Tracing over both legs, $\{t(\lambda), t(\mu)\} = 0$ is the order- $\hbar$ term of [Theorem 17.4](#). With $\lambda + i\hbar P$, false for this chain, the same steps would give $r = +P/\lambda$: the classical side decides the quantum sign. The Dictionary's rows marked T are theorems because of this computation (the check script repeats it for two and three sites).

17.7 The graphical calculus, and the dictionary

The pictures of this lecture are one calculus. A line is a copy of a space: auxiliary lines carry a spectral parameter, site lines carry the chain. A crossing of two lines is an R -matrix (or, with a site line, a Lax operator), a row is a monodromy, a loop is a transfer matrix. Two moves do all the work: *sliding* a line past a crossing, which is the Yang–Baxter equation ([Figure 17.5](#)) or, with a site line, the RLL relation ([Figure 17.4](#)); and *passing a crossing through a row*, the train ([Figure 17.3](#)), which gives [Theorem 17.4](#) ([Figure 17.7](#)). [Figure 8.2](#) was the first crossing, [Figure 8.4](#) the first slide, and [Figure 9.6](#) the same slide with numbers at the crossings; [Lecture 16](#) put matrices there.

Dictionary

The Lax matrix $L(\lambda)$; the periodic Toda site matrix ([10.8](#)) $\rightarrow$ the Lax operator $L_n(\lambda)$: the same matrix with operator entries ([17.13](#)); for spins, ([17.3](#)) (type T, *through quantum Toda*)

The quadratic Sklyanin bracket ([17.1](#)) ([Theorem 8.17](#), [Proposition 10.4](#)) $\rightarrow$ the RTT relation ([17.7](#)) ([Theorem 17.3](#)) (type T: [Corollary 17.10](#))

The classical $r(\lambda) = -P/\lambda$ ([Proposition 10.4](#)) $\rightarrow R(\lambda)/\lambda = \mathbb{1} + i\hbar r(\lambda)$, $R = \lambda - i\hbar P$ (type T, *sign for sign; the ordering is forced*)

$\{t(\lambda), t(\mu)\} = 0$ ([Corollary 8.18](#)) $\rightarrow [t(\lambda), t(\mu)] = 0$ ([Theorem 17.4](#)) (type T: *its $O(\hbar)$ term, through quantum Toda*)

The classical Yang–Baxter equation ([8.13](#)) $\rightarrow$ the Yang–Baxter equation ([17.9](#)) (type T: *the $O(\hbar^2)$ term, reading below*)

The integrals $\text{tr} L^k$ ([Lecture 7](#)); the coefficients of $t(\lambda)$ ([Lecture 10](#)) $\rightarrow$ the local charges $\frac{d^k}{d\lambda^k} \ln t|_0$ of the Heisenberg chain (type A: *the same role; the spin chain has no classical limit here*)

Tutorial

17.8 Two sites by hand

For $N = 2$, $t(\lambda) = \text{tr}_a [(\lambda + iP_{a2})(\lambda + iP_{a1})] = 2\lambda^2 + 2i\lambda - \text{tr}_a (P_{a2}P_{a1})$, using $\text{tr}_a \mathbb{1} = 2$ and $\text{tr}_a P_{an} = 1$. By ([17.4](#)), $P_{a2}P_{a1} = P_{21}P_{a2}$, so $\text{tr}_a (P_{a2}P_{a1}) = P_{12}$ and

$$t(\lambda) = 2\lambda^2 + 2i\lambda - P_{12},$$

which commutes with itself at any two values, as it must: it involves only P_{12} . On the triplet $P_{12} = 1$, on the singlet -1 . Then $t(0) = -P_{12} = i^2 U$ (for $N = 2$ the shift is the swap), and $t(0)^{-1} t'(0) = (-P_{12})(2i) = -2iP_{12}$, which is ([17.11](#)): for $N = 2$ both bonds of the ring are P_{12} . Repeat for $N = 3$ and check $t(0) = -iP_{32}P_{31}$ against [Proposition 17.5](#).

17.9 Reordering three monodromies

Three auxiliary spaces 1, 2, 3 at parameters $\lambda_1, \lambda_2, \lambda_3$ give three relations (17.7), one for each pair, with $R_{ij} = R_{ij}(\lambda_i - \lambda_j)$.

Proposition 17.11 (Reordering three monodromies). *Let T satisfy (17.7) with an invertible R . Reversing $T_1 T_2 T_3$ one neighbouring pair at a time gives, along the two possible routes,*

$$T_1 T_2 T_3 = W_A (T_3 T_2 T_1) W_A^{-1} = W_B (T_3 T_2 T_1) W_B^{-1}, \quad W_A = R_{12} R_{13} R_{23}, \quad W_B = R_{23} R_{13} R_{12}.$$

Hence $W_B^{-1} W_A$ commutes with every $T_3 T_2 T_1$.

Proof. Write (17.7) as $T_i T_j = R_{ij} T_j T_i R_{ij}^{-1}$ for $i < j$. Route A exchanges the pairs 12, then 13, then 23: $T_1 T_2 T_3 = R_{12} T_2 T_1 T_3 R_{12}^{-1} = R_{12} R_{13} T_2 T_3 T_1 R_{13}^{-1} R_{12}^{-1} = W_A T_3 T_2 T_1 W_A^{-1}$, each R_{ij} commuting with the monodromy it does not touch. Route B exchanges 23, 13, 12 and gives W_B in the same way. Equating the two results gives the statement. ■

For a model that satisfies RTT this is automatic — the two routes rewrite one operator. It matters when RTT is taken as the *definition* of an algebra generated by the entries of T , as the Sklyanin bracket was posited in [Lecture 8](#): then the two routes yield the commutation condition of the proposition among the generators, and $W_A = W_B$ makes it automatic. That is sufficient, not necessary: for $R = \sigma^x \otimes \sigma^z$, which fails the Yang–Baxter equation, $W_B = -W_A$ and the condition is again automatic (the check script). It is [Remark 8.13](#) quantised: there the Jacobi identity required a combination of r ’s only to commute with $L_1 + L_2 + L_3$, here the associativity of the products requires $W_B^{-1} W_A$ only to commute with the triple products, and for a particular model that is a weaker demand. What makes the Yang–Baxter equation the right thing to impose is, again, that it mentions no model.

17.10 The Yang–Baxter equation at order $\hbar^2$, and more

$YBE \rightarrow CYBE$. Give each line i a parameter λ_i and write $R_{ij} = R_{ij}(\lambda_i - \lambda_j)$, so (17.9) is $R_{12} R_{13} R_{23} = R_{23} R_{13} R_{12}$. Suppose $R = \mathbb{1} + i\hbar r + \hbar^2 s + O(\hbar^3)$, in difference form. The terms of order $\hbar$ are $r_{12} + r_{13} + r_{23}$ on both sides. Those of order $\hbar^2$ containing s are also the same on both sides; the rest are $(i\hbar)^2$ times the products of two r ’s in the order of the factors, so the difference of the two sides at order $\hbar^2$ is

$$-\hbar^2 \left([r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] \right) :$$

a solution of the Yang–Baxter equation has a classical r that solves the CYBE (8.13), in the per-leg form of § 8.6. For $R = \lambda - i\hbar P$, normalised by $1/\lambda$ (which does not change (17.9), the scalars being the same three on both sides), $r = -P/\lambda$ is [Proposition 8.14](#) at $c = -1$. This is the picture [Figure 8.4](#) promised: the classical slide is the first correction to the quantum one · *proved; the check script repeats it with generic matrices.*

The quantum determinant. At $\lambda - \mu = i$, $R(i) = i(1 - P)$ is $2i$ times the projector onto the antisymmetric part of $\mathbb{C}^2 \otimes \mathbb{C}^2$, and RTT says that $T_a(\mu + i)T_b(\mu)$ maps the antisymmetric line

to itself. The operator by which it multiplies that line (central; for XXX a number) is the *quantum determinant*,

$$\text{qdet } T(\mu) = A(\mu + i)D(\mu) - B(\mu + i)C(\mu),$$

the counterpart of $\det T = 1$ for Toda (10.13). It commutes with every entry of T (Korepin, Bogoliubov and Izergin, §VII.8, in their conventions). For the universal algebra defined by RTT on a normalised formal series $T(u) = \mathbb{1} + T^{(1)}u^{-1} + \dots$, the Yangian $Y(\mathfrak{gl}_2)$, the coefficients of the quantum determinant generate the centre (Molev, ch. 1) · *cited*; a spin chain gives a representation of it. For the Heisenberg chain it is the number $[\mu(\mu + 2i)]^N$ (the check script, $N \leq 3$). [Lecture 18](#) meets its two factors as the eigenvalues of A and D on the reference state.

Other spins. For a spin s at each site, $L_{an}(\lambda) = \lambda + \frac{i}{2} + i\sigma_a \cdot \mathbf{S}_n$ satisfies (17.5) with the same R , by a verification that uses only the commutation relations of $\mathbf{S}$ · *cited*: Korepin, Bogoliubov and Izergin, §VIII.4, (4.6)–(4.8), in their conventions; the check script verifies $s = 1$. The transfer matrix still commutes with itself; but the regular point is lost, since $L(0)$ is no longer a permutation, and a local Hamiltonian needs a transfer matrix with a spin- s auxiliary space (fusion: Kulish, Reshetikhin and Sklyanin 1981; not treated here).

Reading

17.11 Why the zero-curvature equation is not what is quantised

[Lecture 10](#) derived the conservation of $t(\lambda)$ from the zero-curvature equation (10.10), $\dot{L}_n = V_{n+1}L_n - L_nV_n$, and then, separately, the Sklyanin bracket. This lecture quantised only the second. The zero-curvature equation is the equation of motion under one Hamiltonian, and in quantum mechanics equations of motion are not postulated: they follow from $[\cdot, H]$ once H is known. What has to be quantised is the *kinematics*, the brackets, and from them the commuting family and its Hamiltonians follow. The site-by-site transport survives as the monodromy (17.6), and the bracket as RTT . The Heisenberg equation for $T(\lambda)$ under H is then a consequence, and $[H, t(\lambda)] = 0$ is [Theorem 17.4](#), not an input. That is the promise of [Lecture 10](#), and the reason the classical limit of § 17.6 was taken in the bracket, not in the equation of motion.

Summary.

- *The relation:* $R_{ab}(\lambda - \mu)T_b(\mu)T_a(\lambda) = T_a(\lambda)T_b(\mu)R_{ab}(\lambda - \mu)$ with $R(\lambda) = \lambda - i\hbar P$, the ordering forced by $R = \mathbb{1} + i\hbar r$ (§ 17.1). Other books use the other ordering, each with its own R and Lax operator; convert all three before transcribing ([Remark 17.2](#)).
- *Heisenberg chain:* $L_{an}(\lambda) = \lambda + iP_{an}$ satisfies RLL ([Theorem 17.1](#)); it is $-R_{an}(-\lambda)$, and RLL is the Yang–Baxter equation of [Lecture 16](#) (17.9).
- *The train:* RLL at each site implies RTT for $T = L_N \cdots L_1$ ([Theorem 17.3](#)).
- *Commuting transfer matrices:* $[t(\lambda), t(\mu)] = 0$, from RTT and an invertible R only ([Theorem 17.4](#)); the Yang–Baxter equation supplies Lax operators, and is sufficient for the consistency of RTT , not forced by it.
- *The regular point:* $t(0) = i^N U$, the shift ([Proposition 17.5](#)); $\frac{d}{d\lambda} \ln t|_0 = -i \sum P_{n,n+1}$ gives H_{XXX} ([Proposition 17.7](#)); the second derivative is $N - \frac{1}{2}Q_3$ for $N \geq 4$, and the higher ones are local (sketched).

- Quantum Toda: RLL with $\lambda - i\hbar P$ exactly (Theorem 17.8), and at order $\hbar$ the Sklyanin bracket of Lecture 10 with $r = -P/\lambda$, sign for sign (Corollary 17.10).

Tutorial. Exercise Sheet 17.

References

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